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# READhub

*Online Book Platform*

## Final Project Report

Sprint 3 — Complete Submission

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Software Engineering — Agile Development

### Team Members

Fajar | Urwa | Kinza

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# 1. Project Introduction

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## Overview

READhub is a comprehensive online book platform designed to provide users with a seamless experience for discovering, purchasing, renting, and downloading books both physical and digital. It serves as a centralized hub where book enthusiasts can access a wide catalog, manage their purchases, and request books that are not yet available in the system.

The platform caters to two primary user roles: regular users who browse and transact, and administrators who manage inventory, orders, and book requests. Built using modern web technologies with Supabase as the backend infrastructure, READhub ensures secure authentication, real-time data management, and scalable storage for book covers and digital content.

## Problem Statement

Traditional book purchasing and renting is fragmented as users must visit multiple platforms or physical stores to find books, purchase physical copies, or access digital editions. READhub solves this by consolidating all book-related transactions into one platform, with a clean GUI, secure payments, eBook access management, and administrative oversight.

## Project Objectives

- Provide a unified platform for physical book purchase and eBook access (rent/download)
  - Implement a shopping cart system for managing multi-item purchases
  - Allow users to request unavailable books and track request status
  - Equip administrators with tools to manage catalog, orders, and book requests
  - Integrate Supabase for secure, scalable database and file storage
  - Implement a wishlist feature for deferred purchases
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## 2. User Stories (All Sprints)

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### Sprint 1 User Stories

Sprint 1 focused on the foundational features of READhub, establishing core authentication and catalog browsing capabilities.

| Story ID | Title                       | Priority | Status        |
|----------|-----------------------------|----------|---------------|
| US-S1-01 | User Registration           | High     | Done          |
| US-S1-02 | User Login                  | High     | Done          |
| US-S1-03 | Browse Book Catalog         | High     | Done          |
| US-S1-04 | View Book Details           | Medium   | Done          |
| US-S1-05 | Admin: Add/Edit Book        | High     | Done (CO Fix) |
| US-S1-06 | Admin: Inventory Management | High     | Done (CO Fix) |
| US-S1-07 | Search & Filter Books       | Medium   | Done          |

### Sprint 2 User Stories

Sprint 2 built upon the foundation with user profile management, reviews, and admin reporting.

| Story ID | Title                     | Priority | Status        |
|----------|---------------------------|----------|---------------|
| US-S2-01 | Admin: Manage Book Images | High     | Done (CO Fix) |
| US-S2-02 | User Profile Management   | Medium   | Done          |
| US-S2-03 | Book Ratings              | Medium   | Done          |
| US-S2-04 | Book search               | High     | Done (CO Fix) |

### Sprint 3 User Stories (Current Sprint)

#### US-S3-01: Purchase a Physical Book (User)

|             |                                                                                                            |
|-------------|------------------------------------------------------------------------------------------------------------|
| Story ID    | US-S3-01                                                                                                   |
| Story Title | Purchase a Physical Book (User)                                                                            |
| Priority    | High                                                                                                       |
| User Story  | As a registered user, I want to purchase a physical book online so that it can be delivered to my address. |

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### Sub User Stories

| # | Description                                            |
|---|--------------------------------------------------------|
| 1 | Add a physical book to cart                            |
| 2 | Enter delivery address and contact details             |
| 3 | Select payment method (credit card / cash on delivery) |
| 4 | Receive order confirmation with order ID               |
| 5 | View order history                                     |

### Acceptance Criteria

| # | Criteria                                    |
|---|---------------------------------------------|
| 1 | User must be logged in to initiate purchase |
| 2 | Cart displays title, price, quantity        |
| 3 | Delivery form validates required fields     |
| 4 | Payment methods available                   |
| 5 | Unique order ID generated                   |
| 6 | Order confirmation displayed                |
| 7 | Order appears in history                    |
| 8 | Stock decreases after purchase              |

### US-S3-02: Rent or Download an eBook (User)

|             |                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------|
| Story ID    | US-S3-02                                                                                                 |
| Story Title | Rent or Download an eBook (User)                                                                         |
| Priority    | High                                                                                                     |
| User Story  | As a registered user, I want to rent or download an eBook so that I can access digital content flexibly. |

### Sub User Stories

| # | Description                            |
|---|----------------------------------------|
| 1 | Choose rent (7/14/30 days) or download |
| 2 | Pay for access                         |
| 3 | Access eBook from dashboard            |
| 4 | Access revoked after expiry            |

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## Acceptance Criteria

| # | Criteria                        |
|---|---------------------------------|
| 1 | Rent and download options shown |
| 2 | Rental durations available      |
| 3 | Payment required before access  |
| 4 | Downloads appear permanently    |
| 5 | Rentals show expiry date        |
| 6 | Access revoked after expiry     |
| 7 | Download limit max 3 times      |

## US-S3-03: Manage Shopping Cart (User)

|             |                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------|
| Story ID    | US-S3-03                                                                                                              |
| Story Title | Manage Shopping Cart (User)                                                                                           |
| Priority    | High                                                                                                                  |
| User Story  | As a registered user, I want to manage a shopping cart so that I can review and modify my selections before checkout. |

## Sub User Stories

| # | Description             |
|---|-------------------------|
| 1 | Add books to cart       |
| 2 | Update quantity         |
| 3 | Remove books            |
| 4 | View subtotal and total |

## Acceptance Criteria

| # | Criteria                        |
|---|---------------------------------|
| 1 | Add to cart from product pages  |
| 2 | Display title, price, quantity  |
| 3 | Quantity adjustable             |
| 4 | Dynamic total calculation       |
| 5 | Cart saved in database          |
| 6 | Out-of-stock items blocked      |
| 7 | Checkout disabled if cart empty |

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### US-S3-04: Request an Unavailable Book (User)

|             |                                                                                                          |
|-------------|----------------------------------------------------------------------------------------------------------|
| Story ID    | US-S3-04                                                                                                 |
| Story Title | Request an Unavailable Book (User)                                                                       |
| Priority    | Medium                                                                                                   |
| User Story  | As a user, I want to request books that are unavailable so I can be notified when they become available. |

#### Sub User Stories

| # | Description                               |
|---|-------------------------------------------|
| 1 | Fill request form (title, author, reason) |
| 2 | Receive confirmation message              |
| 3 | Track request status in dashboard         |

#### Acceptance Criteria

| # | Criteria                                         |
|---|--------------------------------------------------|
| 1 | Request option available on product/search pages |
| 2 | All form fields validated as required            |
| 3 | Success confirmation shown                       |
| 4 | Duplicate requests blocked                       |

### US-S3-05: Manage Orders (Admin)

|             |                                                                                                   |
|-------------|---------------------------------------------------------------------------------------------------|
| Story ID    | US-S3-05                                                                                          |
| Story Title | Manage Orders (Admin)                                                                             |
| Priority    | High                                                                                              |
| User Story  | As an admin, I want to manage all customer orders so I can process shipments and update statuses. |

#### Sub User Stories

| # | Description                |
|---|----------------------------|
| 1 | View all orders in a table |
| 2 | Filter orders by status    |
| 3 | Update order status        |
| 4 | View order details         |
| 5 | Cancel orders if needed    |

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## Acceptance Criteria

| # | Criteria                                |
|---|-----------------------------------------|
| 1 | Orders table displayed with all records |
| 2 | Status filter options available         |
| 3 | Status update persists to database      |
| 4 | Full order details visible              |
| 5 | Cancel action updates inventory totals  |
| 6 | Delivered orders are not editable       |

## US-S3-06: Manage Book Requests (Admin)

|             |                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------|
| Story ID    | US-S3-06                                                                                                      |
| Story Title | Manage Book Requests (Admin)                                                                                  |
| Priority    | Medium                                                                                                        |
| User Story  | As an admin, I want to review and respond to user book requests to decide whether to add them to the catalog. |

## Sub User Stories

| # | Description                |
|---|----------------------------|
| 1 | View pending book requests |
| 2 | Approve a request          |
| 3 | Reject with reason         |

## Acceptance Criteria

| # | Criteria                                    |
|---|---------------------------------------------|
| 1 | Requests section exists in admin panel      |
| 2 | Request details clearly shown               |
| 3 | Approve and reject options available        |
| 4 | Rejection requires a reason                 |
| 5 | Approved request linked to catalog workflow |

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### US-S3-07: Wishlist Feature (User)

|             |                                                                                         |
|-------------|-----------------------------------------------------------------------------------------|
| Story ID    | US-S3-07                                                                                |
| Story Title | Wishlist Feature (User)                                                                 |
| Priority    | Medium                                                                                  |
| User Story  | As a user, I want to save books to a wishlist so I can revisit and purchase them later. |

#### Sub User Stories

| # | Description                            |
|---|----------------------------------------|
| 1 | Add book to wishlist from product page |
| 2 | Remove book from wishlist              |
| 3 | View dedicated wishlist page           |
| 4 | Move wishlist item to cart             |

#### Acceptance Criteria

| # | Criteria                              |
|---|---------------------------------------|
| 1 | Add/remove wishlist button functional |
| 2 | Wishlist persists across sessions     |
| 3 | Wishlist page shows all saved books   |
| 4 | Move to cart works correctly          |
| 5 | Duplicate items prevented             |

### US-S3-08: Supabase Integration

|             |                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------|
| Story ID    | US-S3-08                                                                                                              |
| Story Title | Supabase Integration                                                                                                  |
| Priority    | High                                                                                                                  |
| User Story  | As a system, all data must be stored and retrieved through Supabase to ensure consistency, security, and scalability. |

#### Sub User Stories

| # | Description                                  |
|---|----------------------------------------------|
| 1 | Database tables created for all entities     |
| 2 | CRUD operations work via Supabase client     |
| 3 | Book cover images stored in Supabase Storage |
| 4 | Authentication managed through Supabase Auth |

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## Acceptance Criteria

| # | Criteria                                      |
|---|-----------------------------------------------|
| 1 | All data stored in Supabase PostgreSQL tables |
| 2 | CRUD operations work for all modules          |
| 3 | Images stored in and retrieved from Storage   |
| 4 | Auth tokens and sessions managed securely     |
| 5 | Correct query results returned for all views  |

## 3. Sprint Backlog — Sprint 3

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### Sprint Backlog Table

| Story ID   | Task                    | Assigned To | Priority | Status |
|------------|-------------------------|-------------|----------|--------|
| US-S3-01   | Purchase Book           | Fajar       | High     | To Do  |
| US-S3-01.1 | Cart                    | Fajar       | High     | To Do  |
| US-S3-01.2 | Address & Payment       | Urwa        | High     | To Do  |
| US-S3-01.3 | Order History           | Fajar       | High     | To Do  |
| US-S3-02   | eBook Rent/Download     | Urwa        | High     | To Do  |
| US-S3-02.1 | UI Implementation       | Urwa        | High     | To Do  |
| US-S3-02.2 | Expiry Logic            | Kinza       | High     | To Do  |
| US-S3-02.3 | User Dashboard          | Urwa        | Medium   | To Do  |
| US-S3-03   | Cart Management         | Fajar       | High     | To Do  |
| US-S3-03.1 | Cart UI                 | Fajar       | High     | To Do  |
| US-S3-04   | Request Book            | Kinza       | Medium   | To Do  |
| US-S3-04.1 | Request Form            | Kinza       | Medium   | To Do  |
| US-S3-04.2 | Status Tracking         | Kinza       | Medium   | To Do  |
| US-S3-05   | Orders Admin            | Fajar       | High     | To Do  |
| US-S3-05.1 | Orders List             | Fajar       | High     | To Do  |
| US-S3-05.2 | Status Update           | Urwa        | High     | To Do  |
| US-S3-05.3 | Order Details           | Fajar       | High     | To Do  |
| US-S3-06   | Requests Admin          | Kinza       | Medium   | To Do  |
| US-S3-06.1 | Approve/Reject Workflow | Kinza       | Medium   | To Do  |
| US-S3-07   | Wishlist                | Urwa/Kinza  | Medium   | To Do  |
| CO-S1-06   | Inventory Fix           | Fajar       | High     | To Do  |

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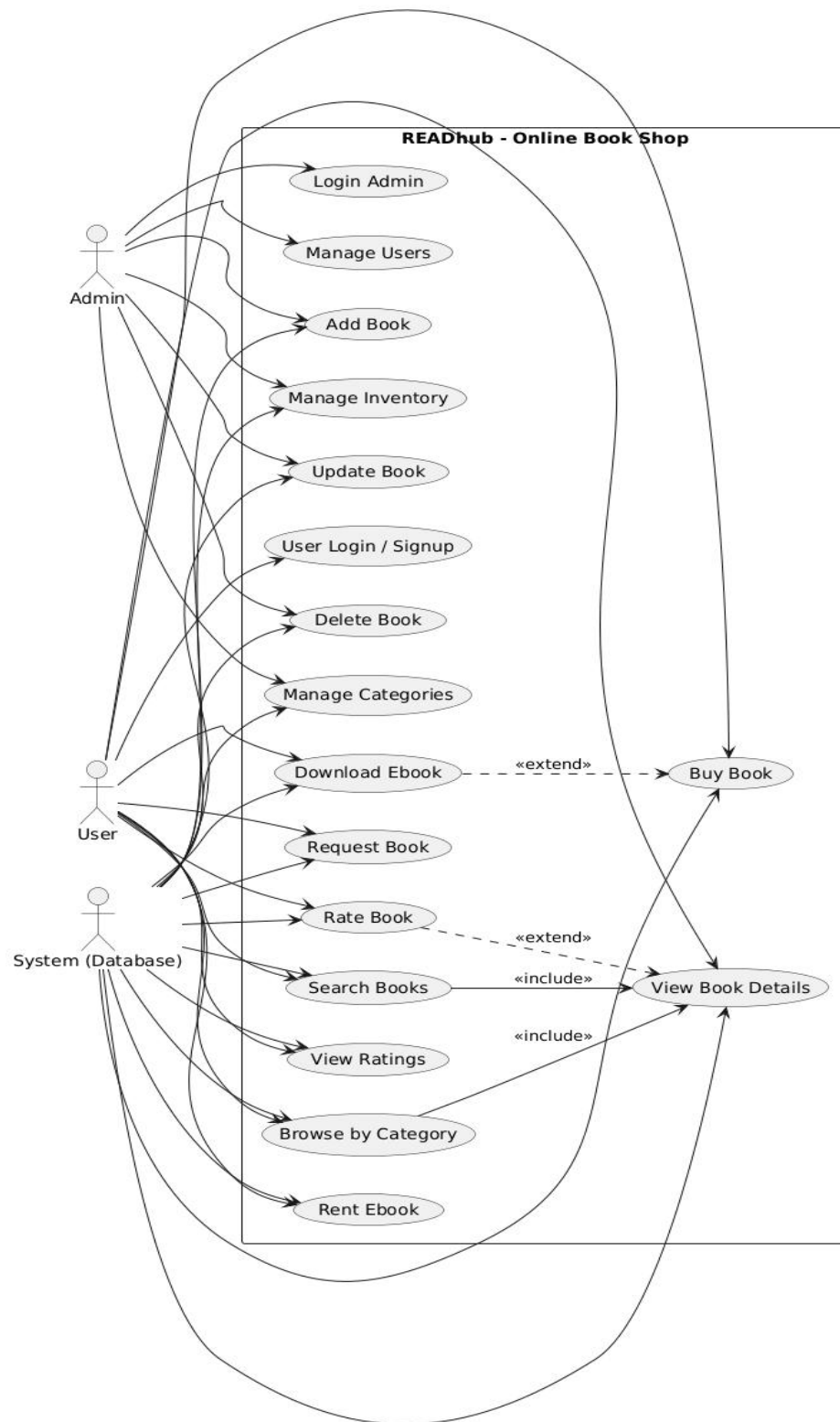
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|            |                   |       |        |       |
|------------|-------------------|-------|--------|-------|
| CO-S1-05.4 | FK Constraint Fix | Fajar | Medium | To Do |
| CO-S2-01.2 | Image Validation  | Urwa  | High   | To Do |
| CO-S2-01.3 | Category Fix      | Urwa  | High   | To Do |
| CO-S2-06.2 | Rating Fix        | Urwa  | High   | To Do |

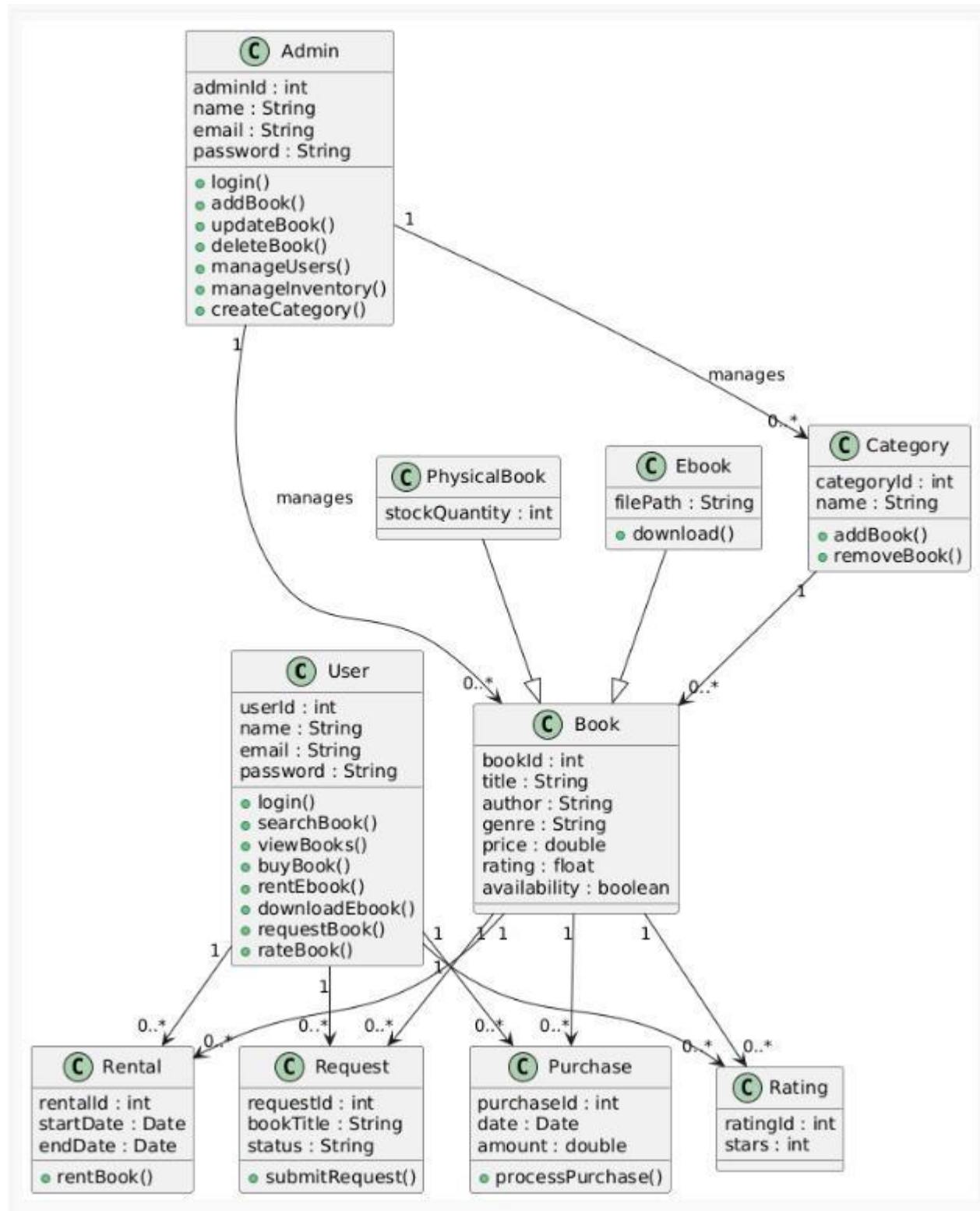
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## 4. Design

### Use Cse Diagram



## Class Diagram Overview

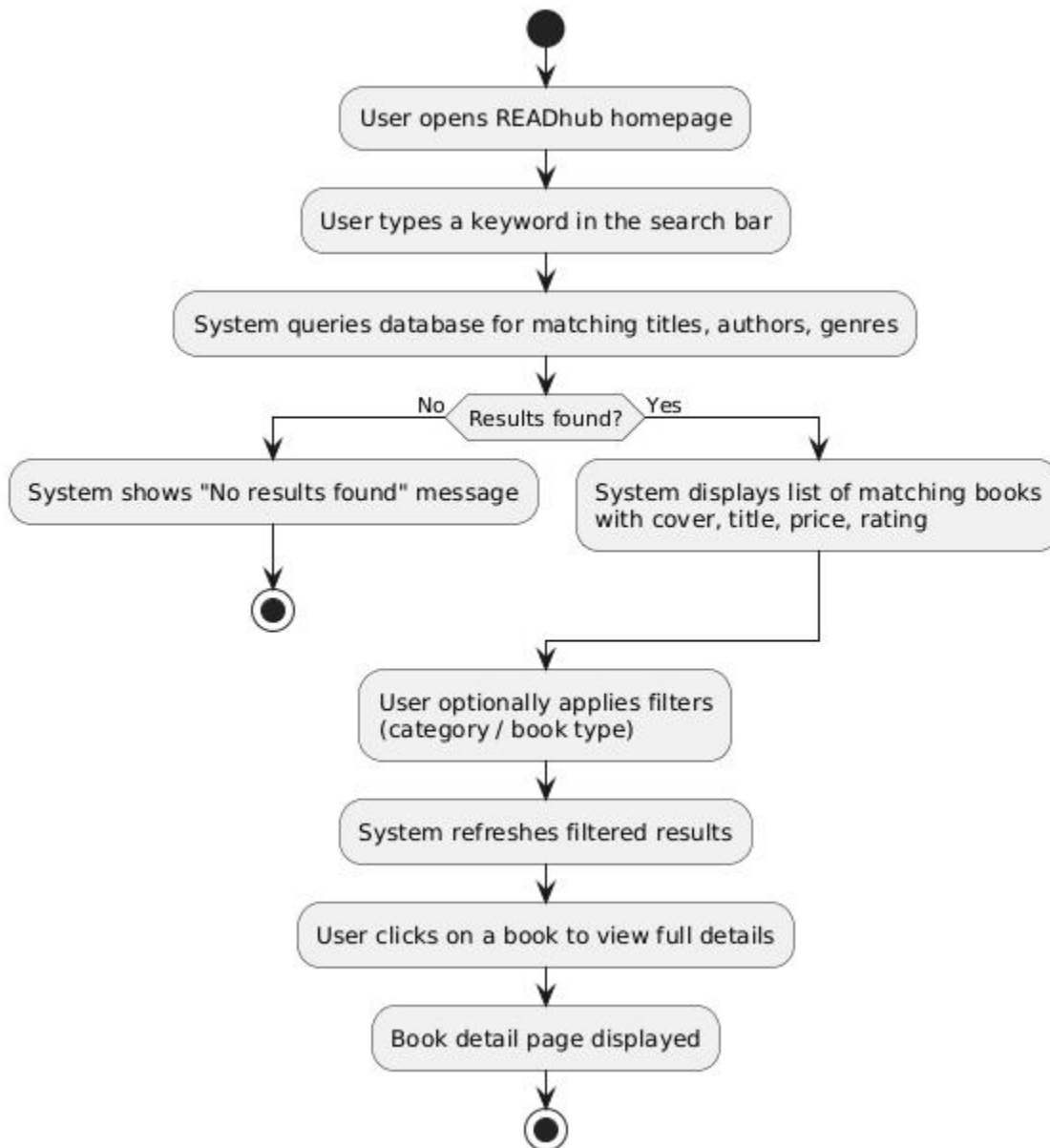


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## Activity Diagram

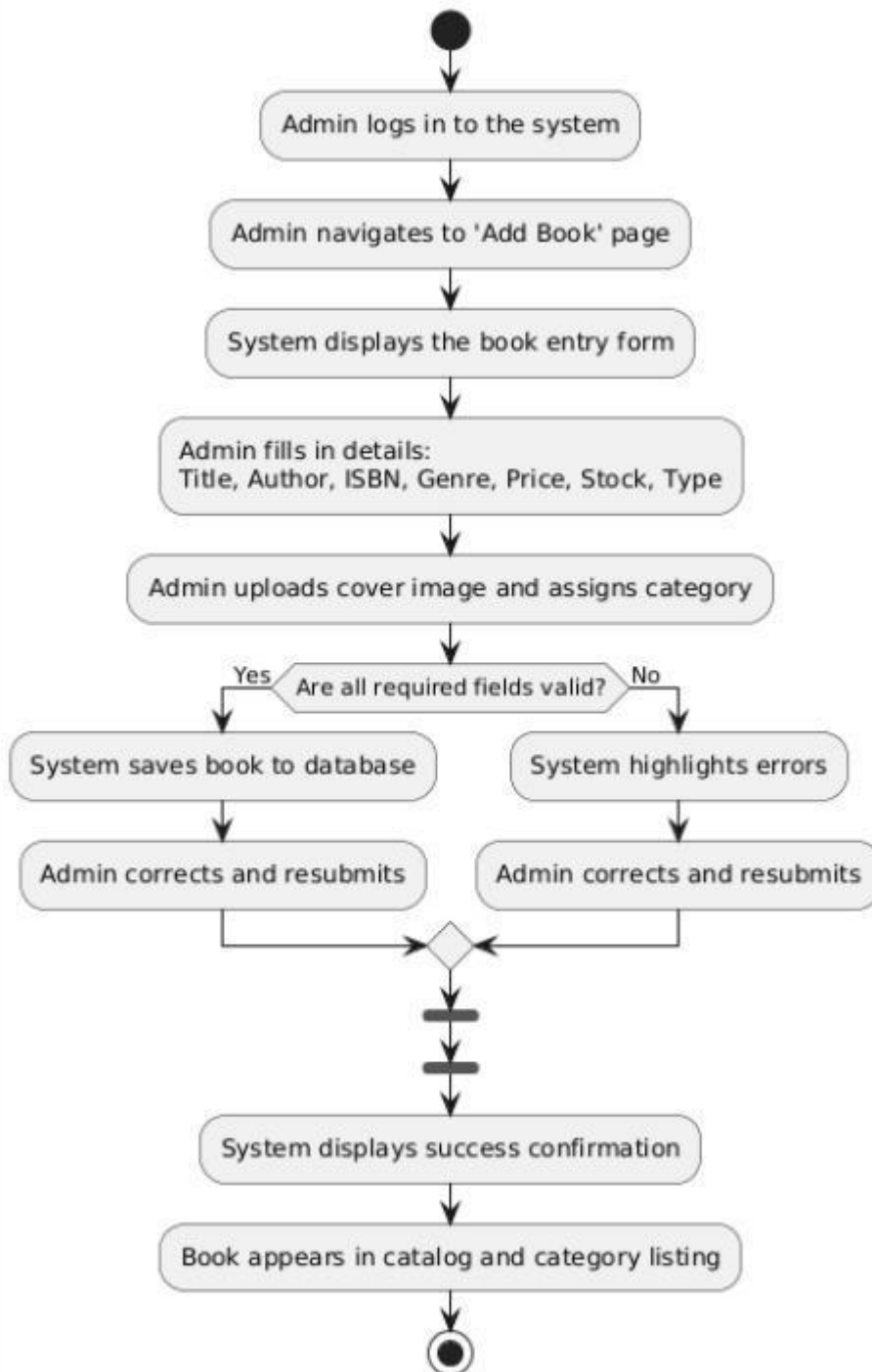
Activity diagrams illustrate the workflow and step-by-step processes within the Book Management System. The three main activities selected correspond directly to the primary user stories of this sprint.

### Activity Diagram 1 — Search for a Book (User)



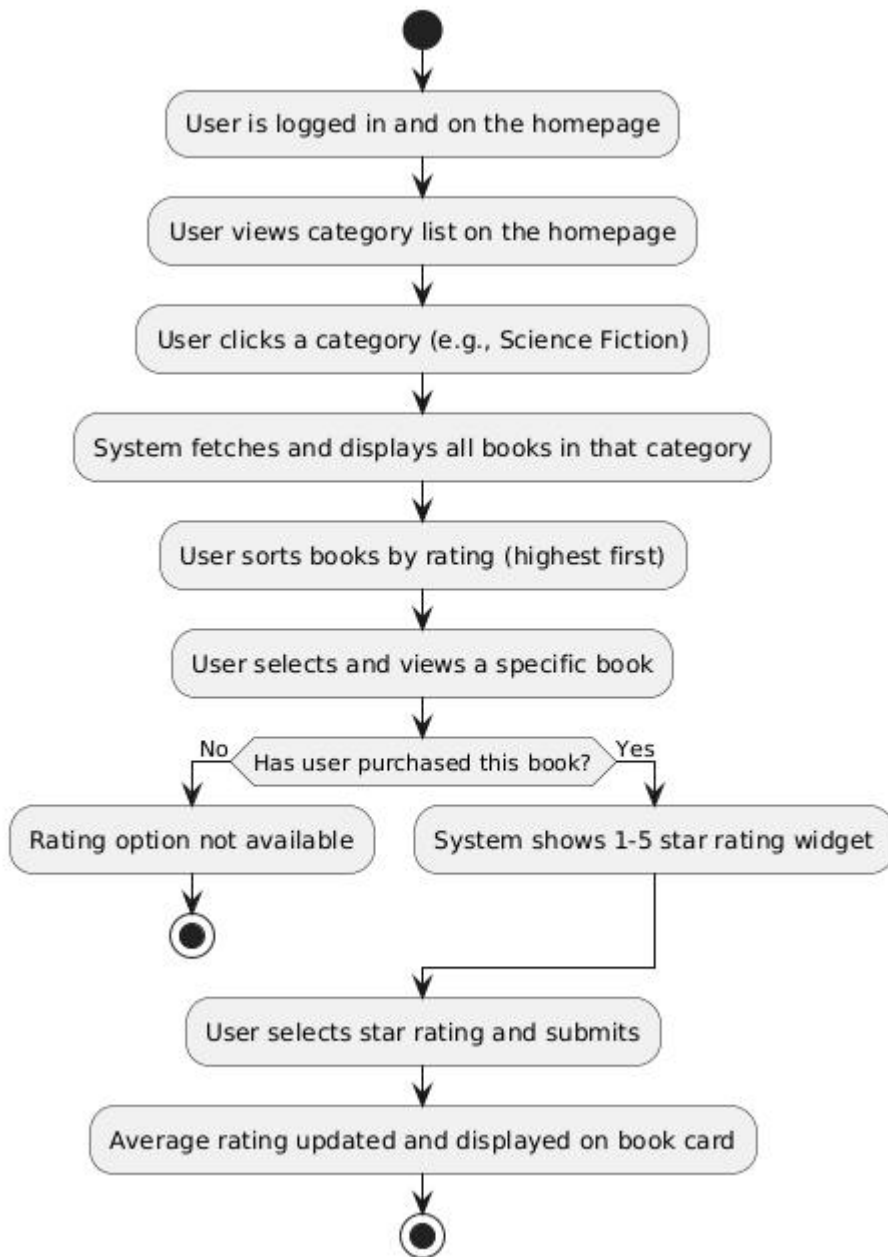
## Activity Diagram 2 — Add New Book (Admin)

### Activity Diagram 2 — Add New Book (Admin)



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### Activity Diagram 3 — Browse by Category and Rate a Book (User)



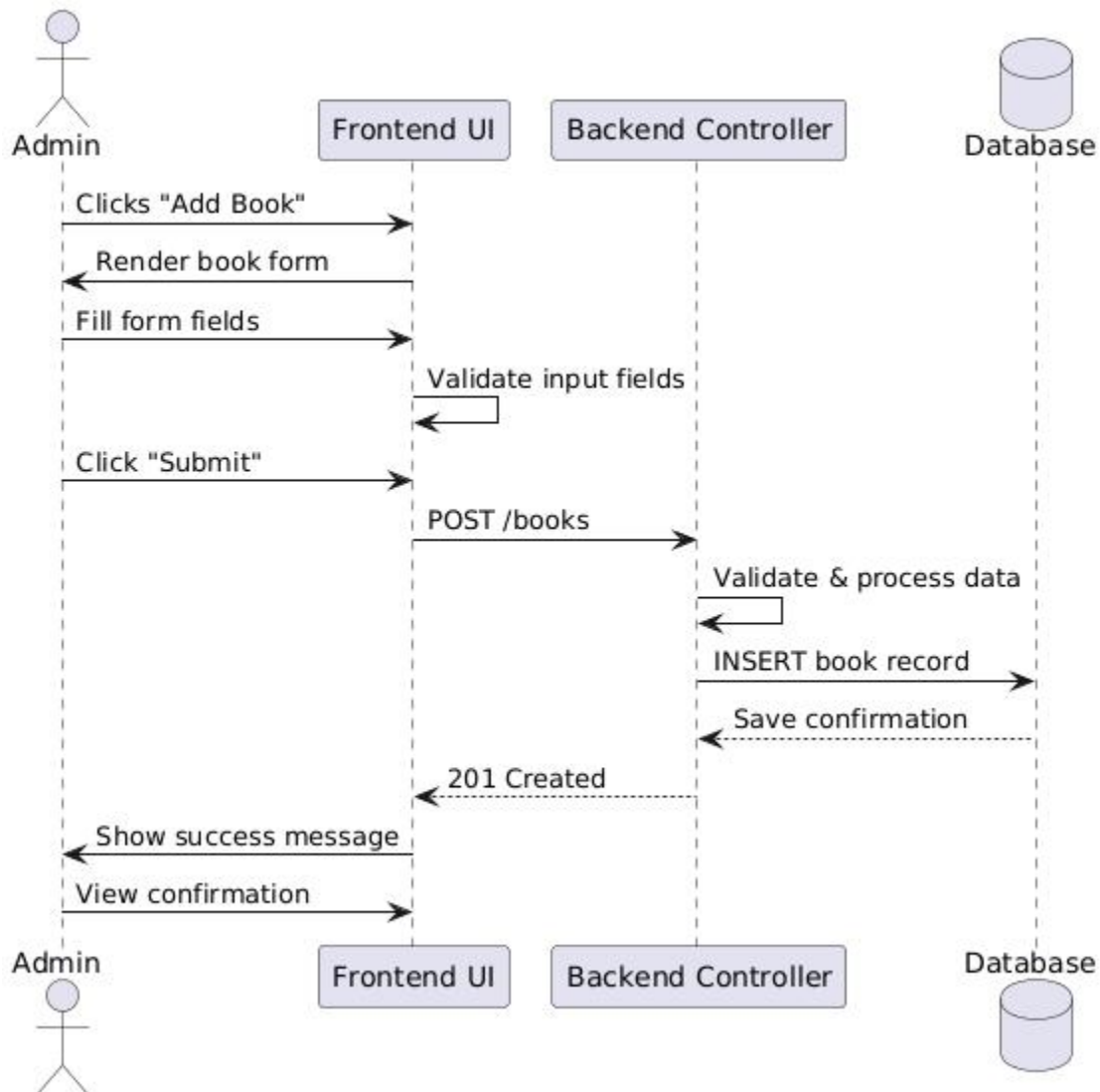


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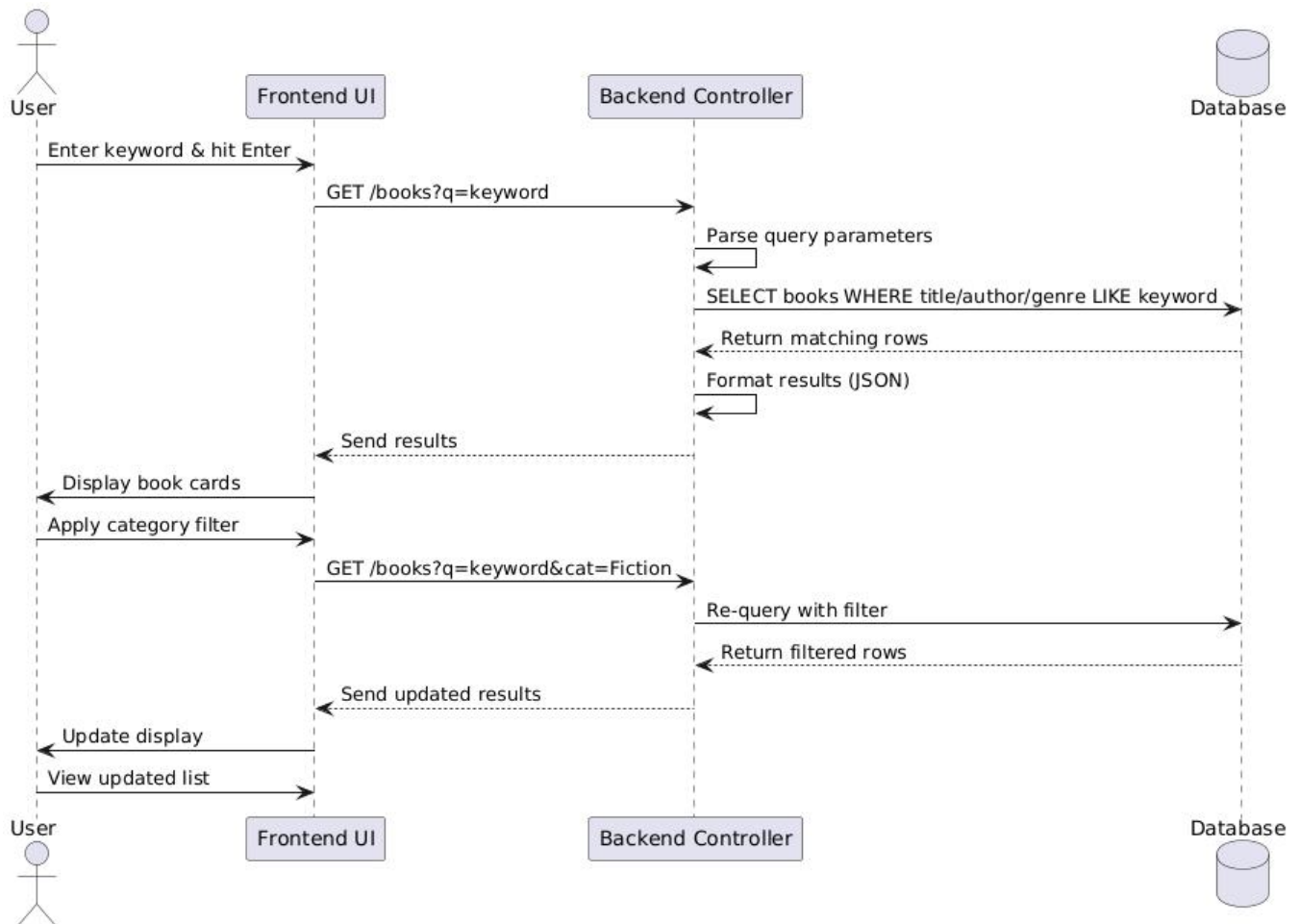
## Sequence Diagram

Sequence diagrams show the order of interactions between actors and system components for the three main processes.

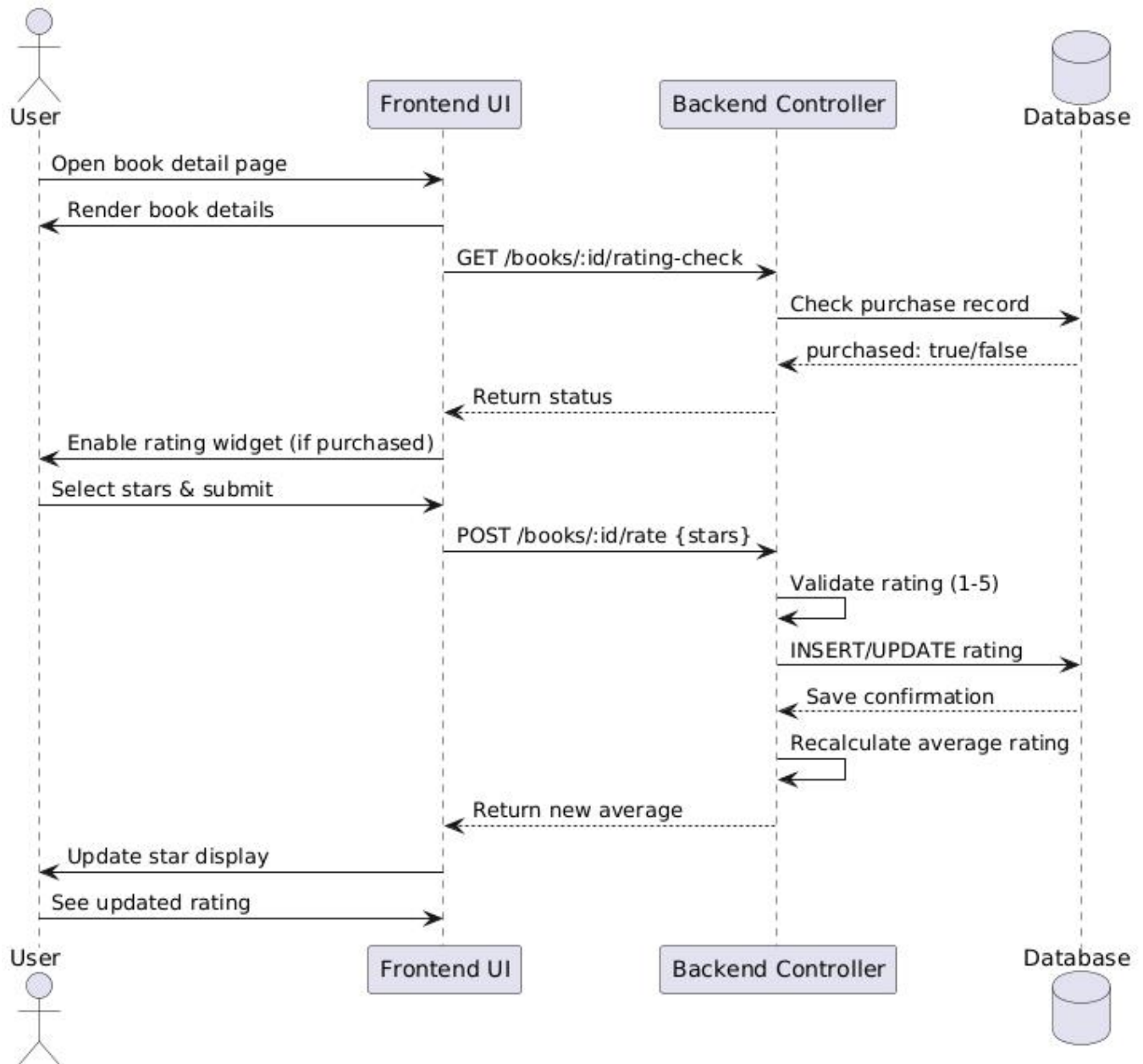
### Sequence Diagram 1 — Add New Book



## Sequence Diagram 2 — Search for a Book



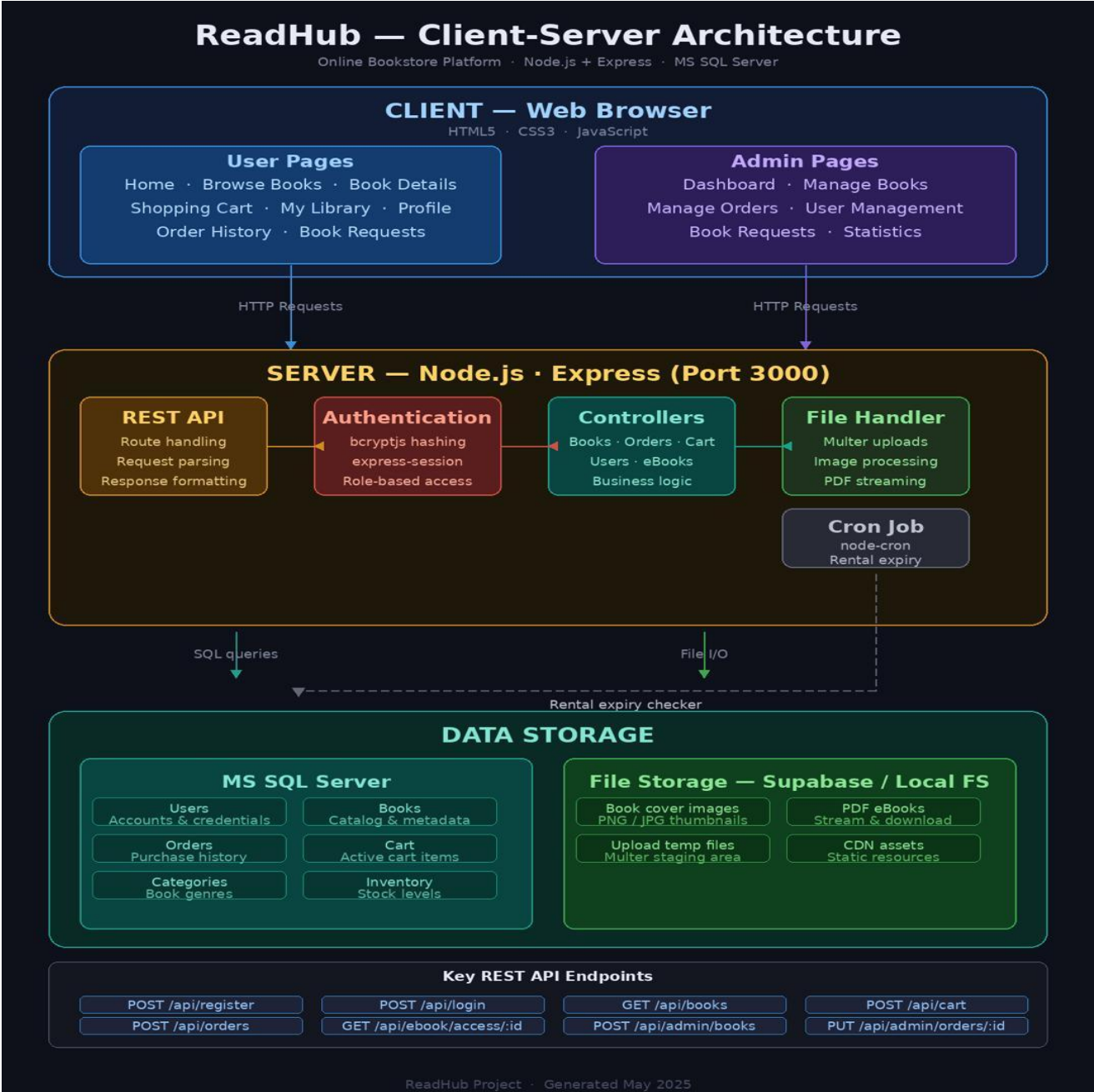
### Sequence Diagram 3 — Rate a Book



# 5. Architecture

## Architecture Pattern: Client-Server with MVC

READhub follows a Client-Server architecture pattern combined with the Model-View-Controller (MVC) design pattern. This ensures a clear separation of concerns between the presentation layer, business logic, and data management.



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## Non-Functional Requirements

### NFR-01: Security

The READhub system ensures a high level of security across user authentication, data access, and digital content protection. User authentication is managed through Supabase Auth, where all passwords are securely hashed using bcrypt before storage. Session management is handled with tokens with controlled expiration to reduce the risk of unauthorized access.

### NFR-02: Performance

**Response Time:** The application backend should process and respond to API requests (like fetching contracts, tokens, or books) within 200-500 milliseconds under normal load.

**Page Load Time:** The frontend (React Application Shell / Admin Dashboard) should load the initial layout in under 2 seconds. Client-side routing should switch between views almost instantly without a full page reload.

**Concurrent Users:** The system must handle multiple concurrent users (e.g., hundreds of clients and freelancers) without significant degradation in response time.

**Scalability:** The architecture (separating frontend and backend Node.js APIs) must support horizontal scaling, allowing more backend instances to be added if traffic increases.

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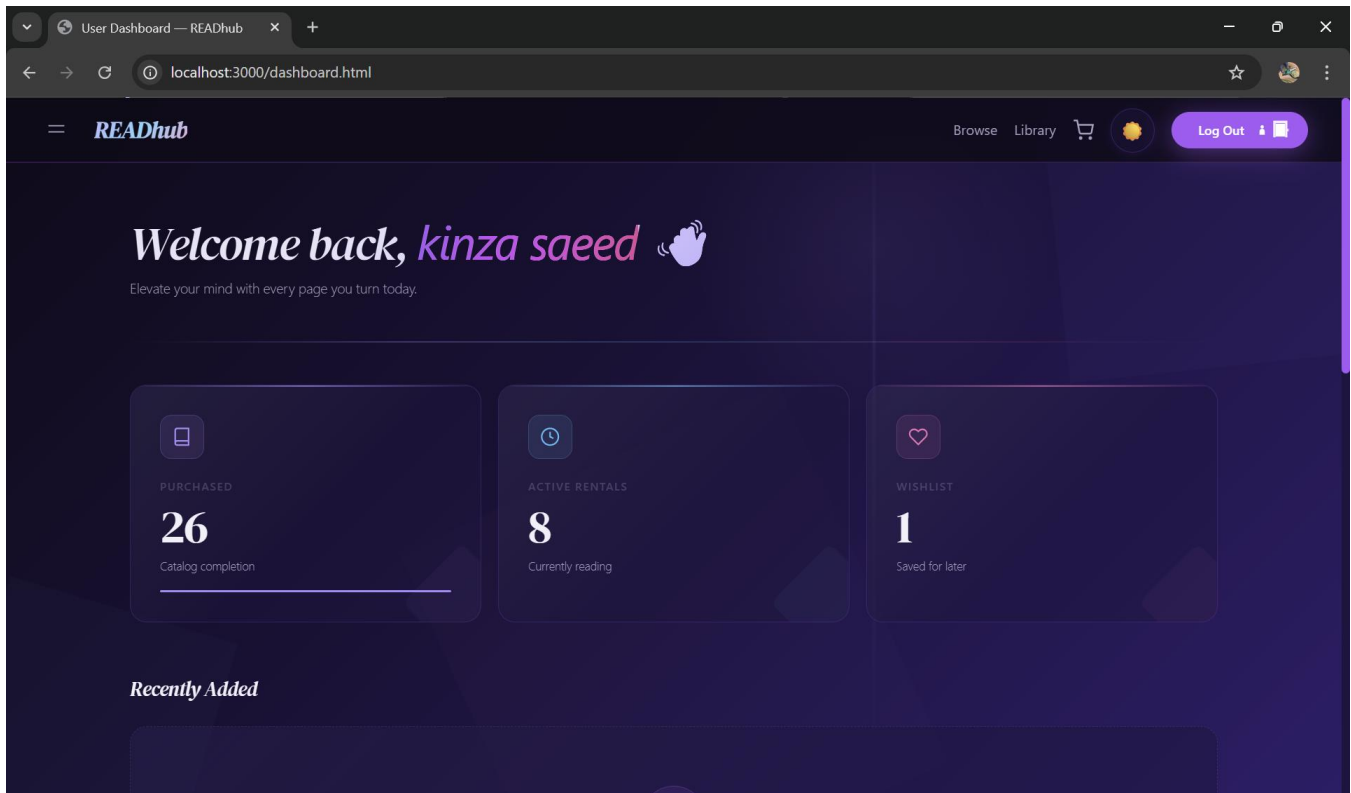
## 6. Implementation Screenshots

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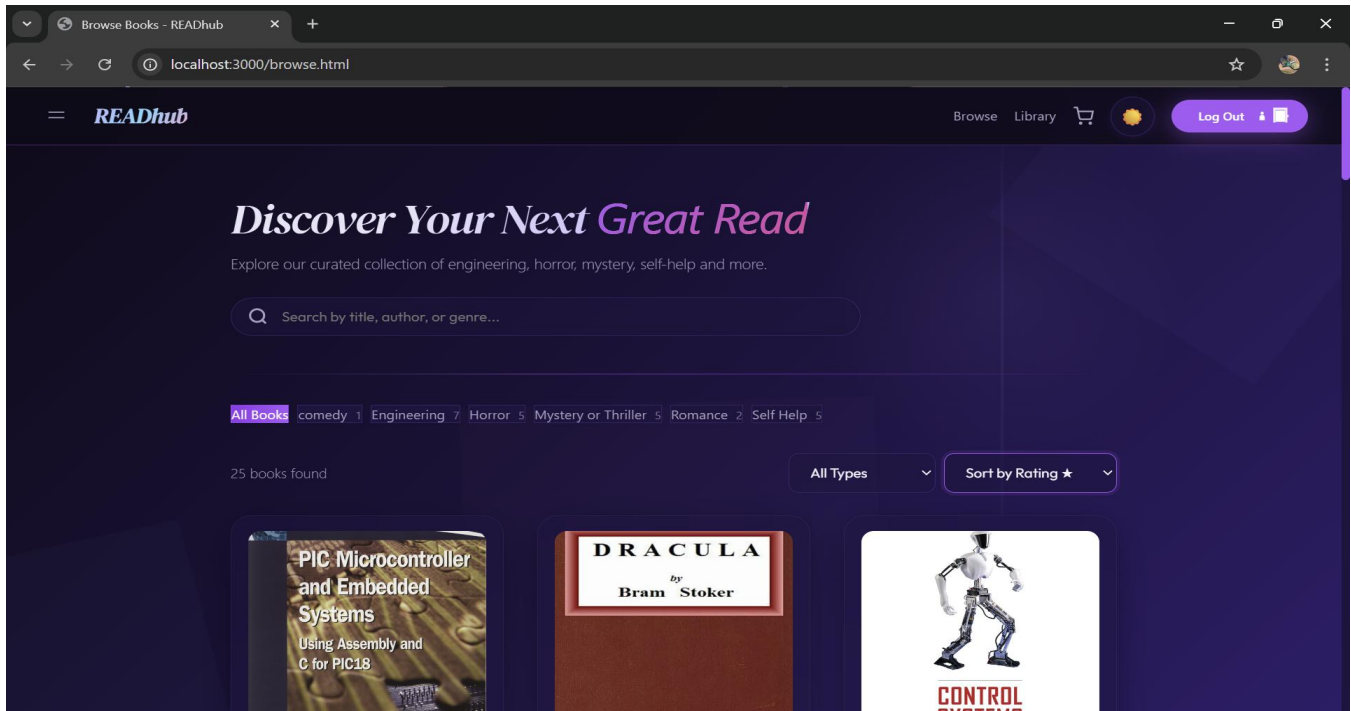
The following section describes the key implemented screens of READhub. Screenshots from the actual running application are to be inserted in this section during final submission.

### 6.1 User-Facing Screens

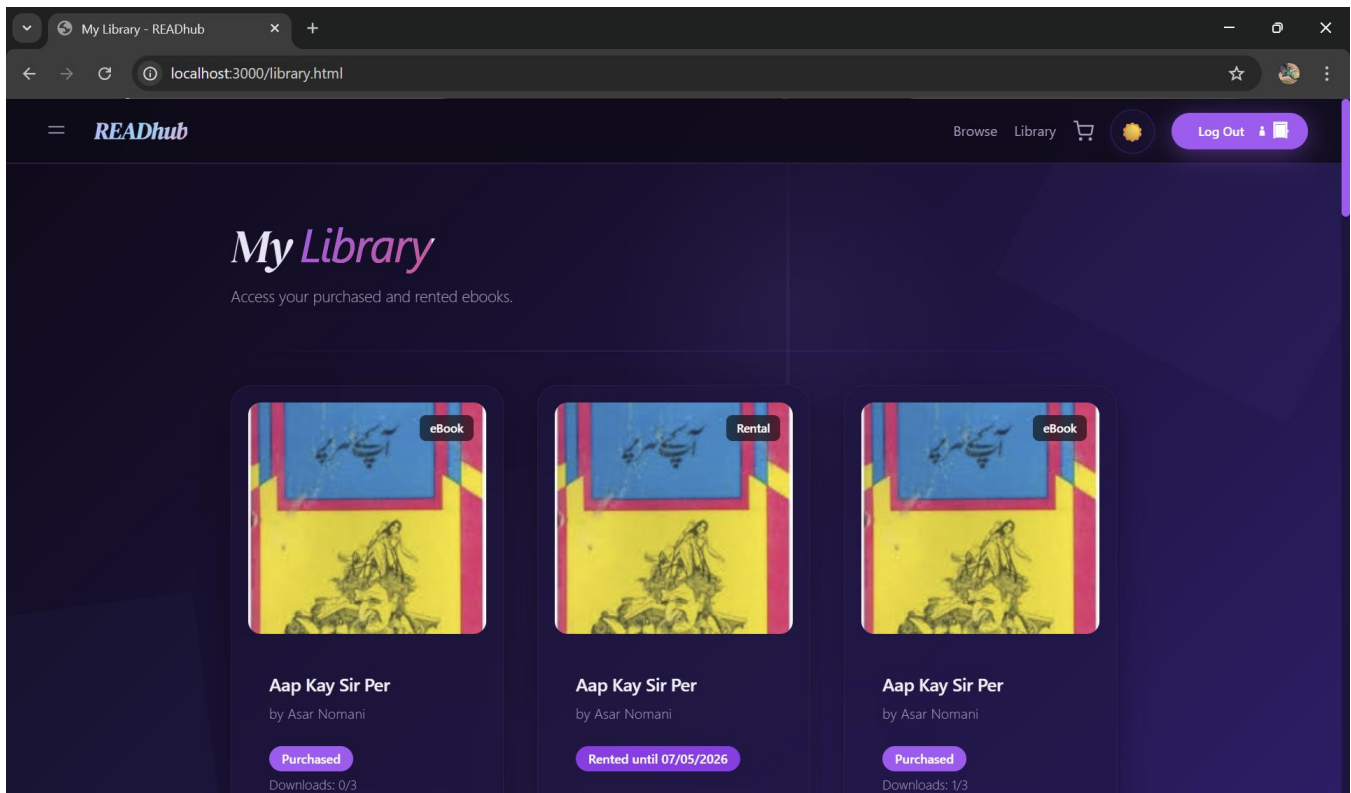
#### 6.1.1 User Dashboard



## 6.1.2 Browse Book Page



## 6.1.3 User Library



## 6.1.4 Book Description Page

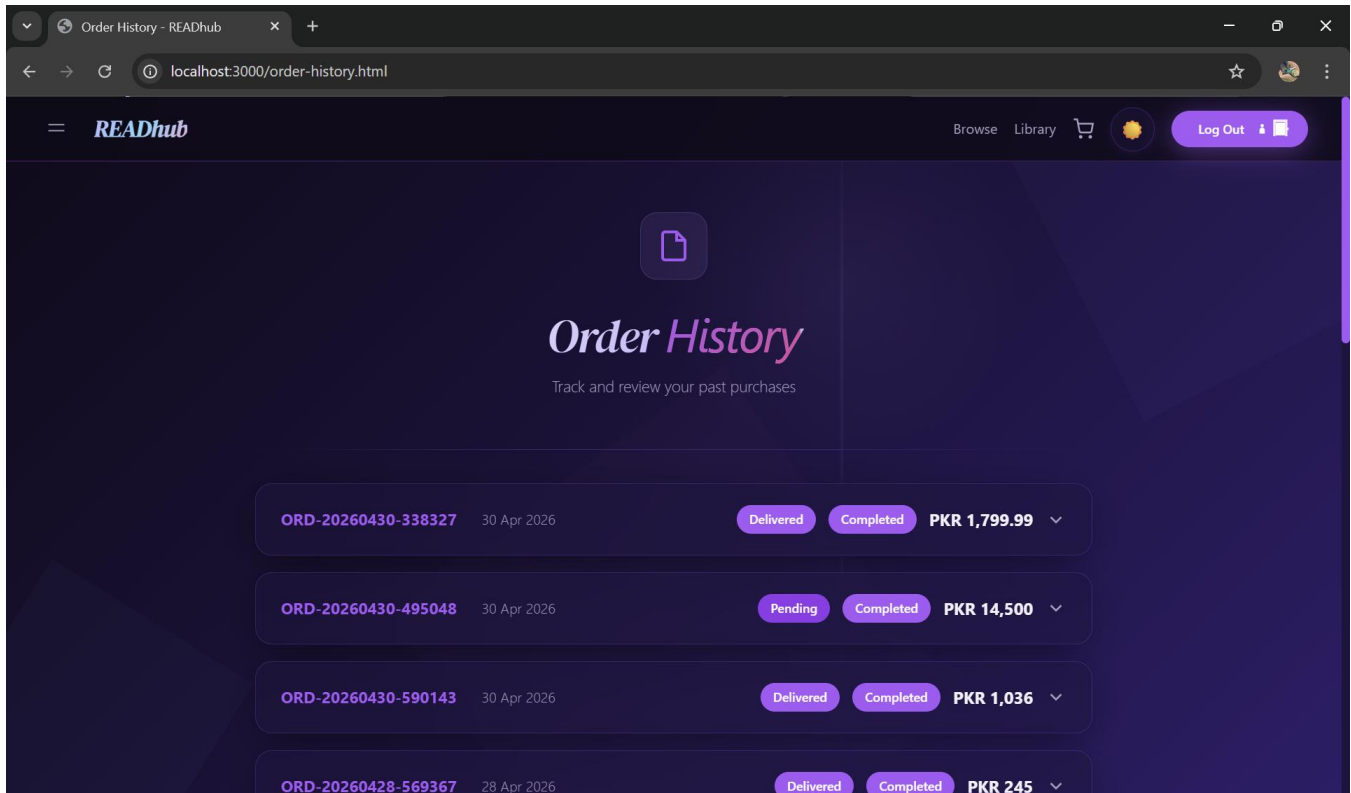
The screenshot shows a web browser window with the URL `localhost:3000/book-details.html?id=4`. The page features a dark blue header with the READhub logo, navigation links (Browse, Library), and a shopping cart icon. The main content area displays the book cover for 'Fundamentals of Thermodynamics' (Tenth Edition) by Claus Borgnakke and Richard E. Sonntag. The book cover shows a blue background with red and yellow text and a technical illustration of a pipe system. To the right of the cover, the book title is prominently displayed, followed by the authors' names. Below this, a rating section shows a 0.0 (Aggregate Rating) with five empty star icons. The category is listed as 'Engineering' and the status is 'In Stock'. A description box follows, stating: 'This leading textbook provides a comprehensive and rigorous introduction to the essential principles of classical thermodynamics within an engineering context. It is designed to prepare students for the practical application of thermodynamics in engineering and to lay the foundation for advanced studies in fluid mechanics and heat transfer. The tenth edition features updated data, enhanced end-of-chapter problems, and a new chapter on...'.

## 6.1.5 User Cart

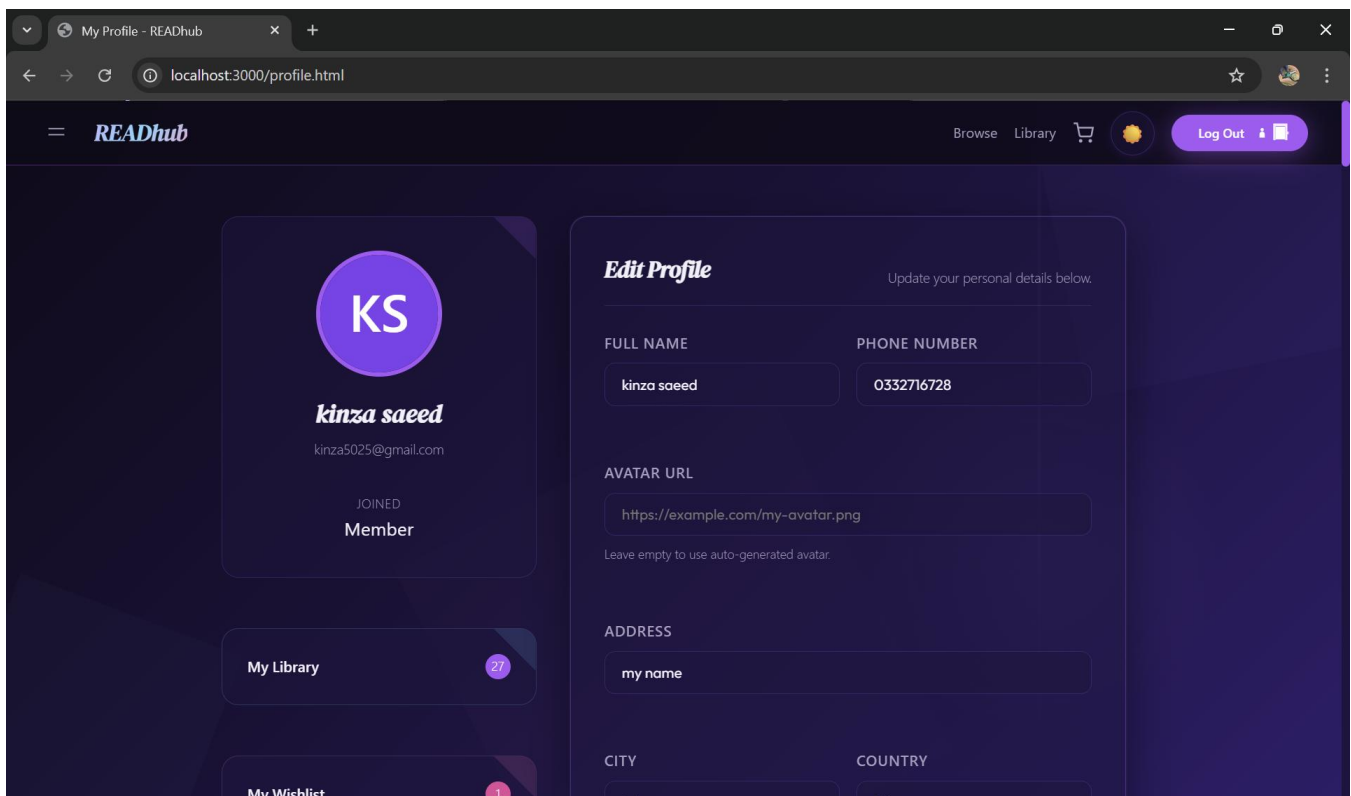
The screenshot shows a web browser window with the URL `localhost:3000/cart.html`. The page features a dark blue header with the READhub logo, navigation links (Browse, Library), and a shopping cart icon. The main content area displays the 'Shopping Cart' title and a subtitle 'Review your items before checkout'. Below this, a single item is listed: 'Fundamentals of Thermodynamics' by Claus Borgnakke and Richard E. Sonntag. The item is shown with its cover image, a quantity of 1, and a price of PKR 1,799.99. A button labeled 'eBook' is visible next to the item. To the right, an 'Order Summary' box contains the following information: 'Items (1)' with a price of PKR 1,799.99, and a 'Total' of PKR 1,799.99. Below the summary, there are three buttons: 'Proceed to Checkout' (highlighted in red), 'Clear Cart', and 'Continue Shopping' (with a left arrow).



## 6.1.6 Order History Page



## 6.1.7 User Profile



## 6.1.8 Book Request Form

Request a Book - READhub

localhost:3000/book-request.html

READhub

BrowseLibrary

Log Out

# Request a New Book

Can't find what you're looking for? Let us know!

### Submit a Request

BOOK TITLE \*

Enter book title

AUTHOR NAME \*

Enter author name

ADDITIONAL NOTES (OPTIONAL)

Any specific edition or details?

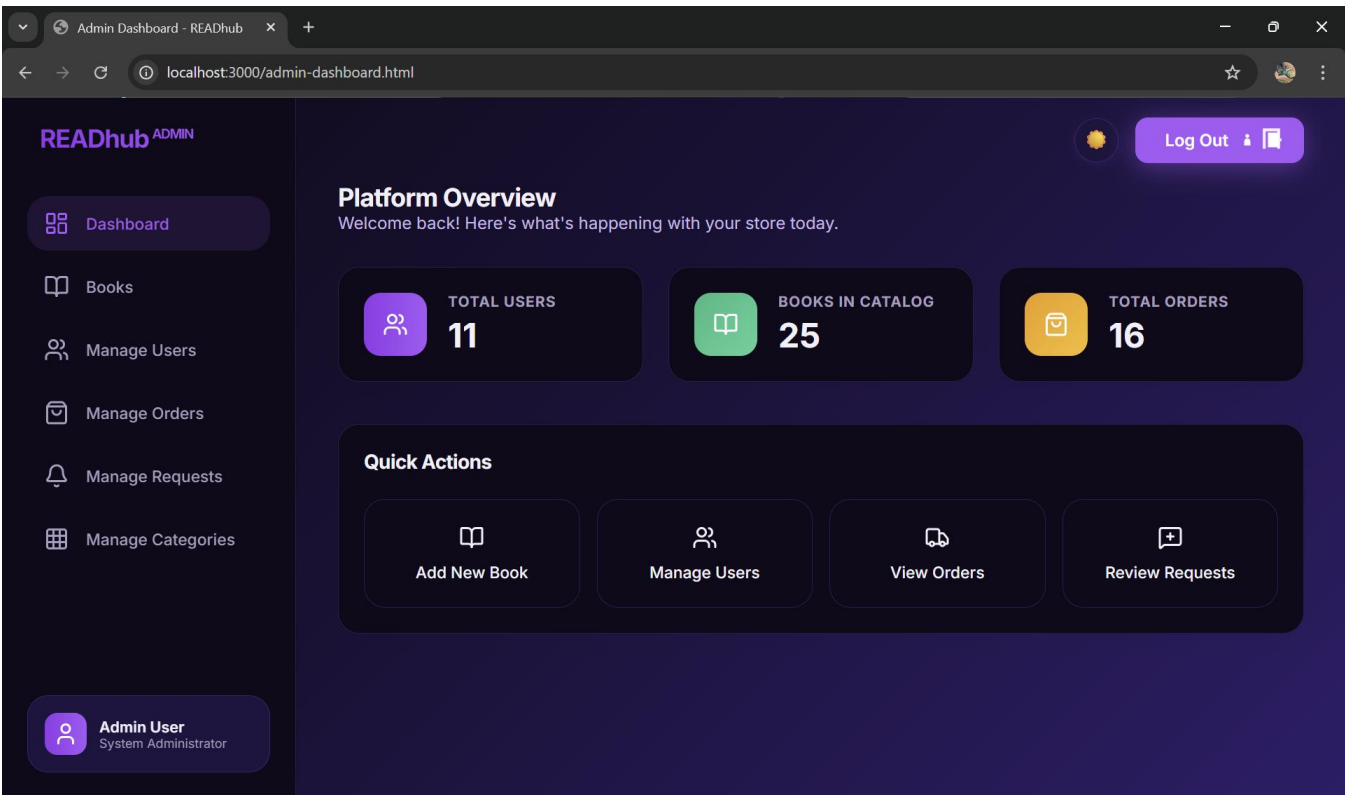
### My Previous Requests

Track the status of your book requests here.

| BOOK TITLE | AUTHOR | DATE       | STATUS   |
|------------|--------|------------|----------|
| dnsj       | fgh    | 25/04/2026 | Reviewed |
| gwud       | hquah  | 25/04/2026 | Reviewed |
| new        | rig    | 25/04/2026 | Pending  |

## 6.2 Admin Panel Screens

### 6.2.1 Admin Dashboard



The screenshot shows the READhub Admin Dashboard. The left sidebar contains navigation links: Dashboard, Books, Manage Users, Manage Orders, Manage Requests, and Manage Categories. The main content area is titled "Platform Overview" and includes a welcome message. It features three summary cards: "TOTAL USERS" (11), "BOOKS IN CATALOG" (25), and "TOTAL ORDERS" (16). Below these is a "Quick Actions" section with buttons for "Add New Book", "Manage Users", "View Orders", and "Review Requests". The bottom left shows the user profile for "Admin User" (System Administrator). The top right has a "Log Out" button.

**Platform Overview**  
Welcome back! Here's what's happening with your store today.

**Summary Cards:**

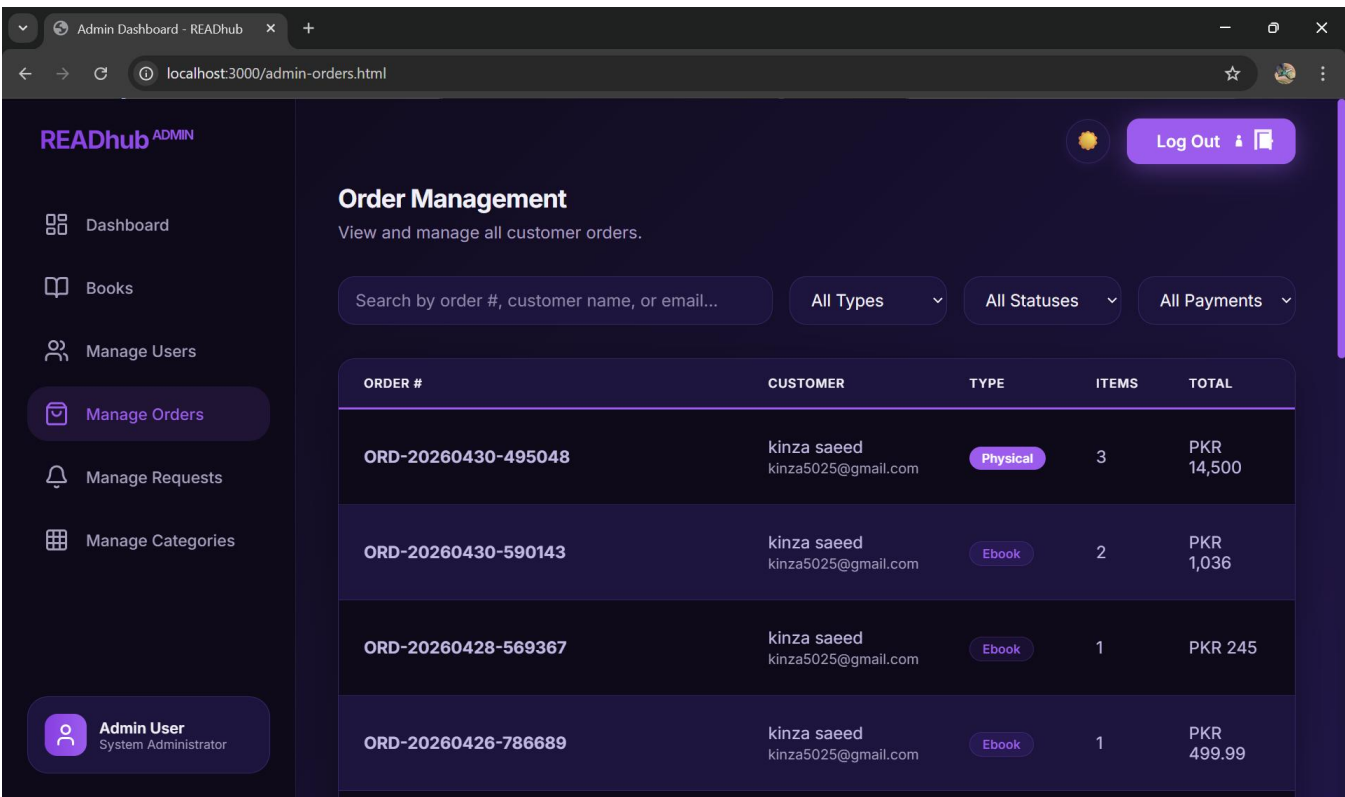
- TOTAL USERS: 11
- BOOKS IN CATALOG: 25
- TOTAL ORDERS: 16

**Quick Actions:**

- Add New Book
- Manage Users
- View Orders
- Review Requests

**User Profile:** Admin User (System Administrator)

### 6.2.2 Order Management



The screenshot shows the READhub Admin Order Management screen. The left sidebar is the same as the dashboard. The main content area is titled "Order Management" and includes a subtitle "View and manage all customer orders." There are three filter buttons: "Search by order #, customer name, or email...", "All Types", "All Statuses", and "All Payments". Below the filters is a table with four columns: ORDER #, CUSTOMER, TYPE, ITEMS, and TOTAL. The table contains four rows of order data.

**Order Management**  
View and manage all customer orders.

**Filters:**

- Search by order #, customer name, or email...
- All Types
- All Statuses
- All Payments

| ORDER #             | CUSTOMER                           | TYPE     | ITEMS | TOTAL      |
|---------------------|------------------------------------|----------|-------|------------|
| ORD-20260430-495048 | kinza saeed<br>kinza5025@gmail.com | Physical | 3     | PKR 14,500 |
| ORD-20260430-590143 | kinza saeed<br>kinza5025@gmail.com | Ebook    | 2     | PKR 1,036  |
| ORD-20260428-569367 | kinza saeed<br>kinza5025@gmail.com | Ebook    | 1     | PKR 245    |
| ORD-20260426-786689 | kinza saeed<br>kinza5025@gmail.com | Ebook    | 1     | PKR 499.99 |

### 6.2.3 Books Management

READhubADMIN

Dashboard

Books

Manage Users

Manage Orders

Manage Requests

Manage Categories

Admin User

System Administrator

Book Catalog Management

Add, edit, and remove books from the catalog.

Search by title, author, or ISBN...

All Categories

All Types

| COVER | TITLE                                                                              | AUTHOR               | CATEGORY | TYPE              | PRICE      | STOCK |
|-------|------------------------------------------------------------------------------------|----------------------|----------|-------------------|------------|-------|
|       | Aap Kay Sir Per                                                                    | Asar Nomani          | comedy   | Physical<br>eBook | PKR 13,464 | 54    |
|       | The Valley Of Silent Men A Story Of The Three River Country<br>978-969-37-019-0678 | James Oliver Curwood | Romance  | Physical<br>eBook | PKR 8,907  | 88    |
|       | Marked<br><a href="#">View PDF</a>                                                 | Chantol C. Aspinall  | Romance  | Physical<br>eBook | PKR 2,400  | 0     |

### 6.2.4 User Management

READhubADMIN

Dashboard

Books

Manage Users

Manage Orders

Manage Requests

Manage Categories

Admin User

System Administrator

User Management

View, search, and manage all registered accounts.

Search by name, email, or username...

All Roles

Sort by ID

| ID  | FULL NAME   | EMAIL                 | USERNAME              | ROLE     | STATUS |
|-----|-------------|-----------------------|-----------------------|----------|--------|
| #1  | Ahmad Saeed | l242527@lhr.nu.edu.pk | l242527@lhr.nu.edu.pk | Customer | Active |
| #10 | Test User   | test@gmail.com        | test@gmail.com        | Customer | Active |
| #11 | Test User   | testuser@gmail.com    | testuser@gmail.com    | Customer | Active |
| #2  | kinza saeed | kinza5025@gmail.com   | kinza5025@gmail.com   | Customer | Active |
| #3  | Admin User  | admin@gmail.com       | admin1                | Admin    | Active |

## 6.2.5 Book request Management

Admin Dashboard - READhub

localhost:3000/admin-book-requests.html

READhubADMIN

Dashboard

Books

Manage Users

Manage Orders

Manage Requests

Manage Categories

Admin User

System Administrator

Book Requests Management

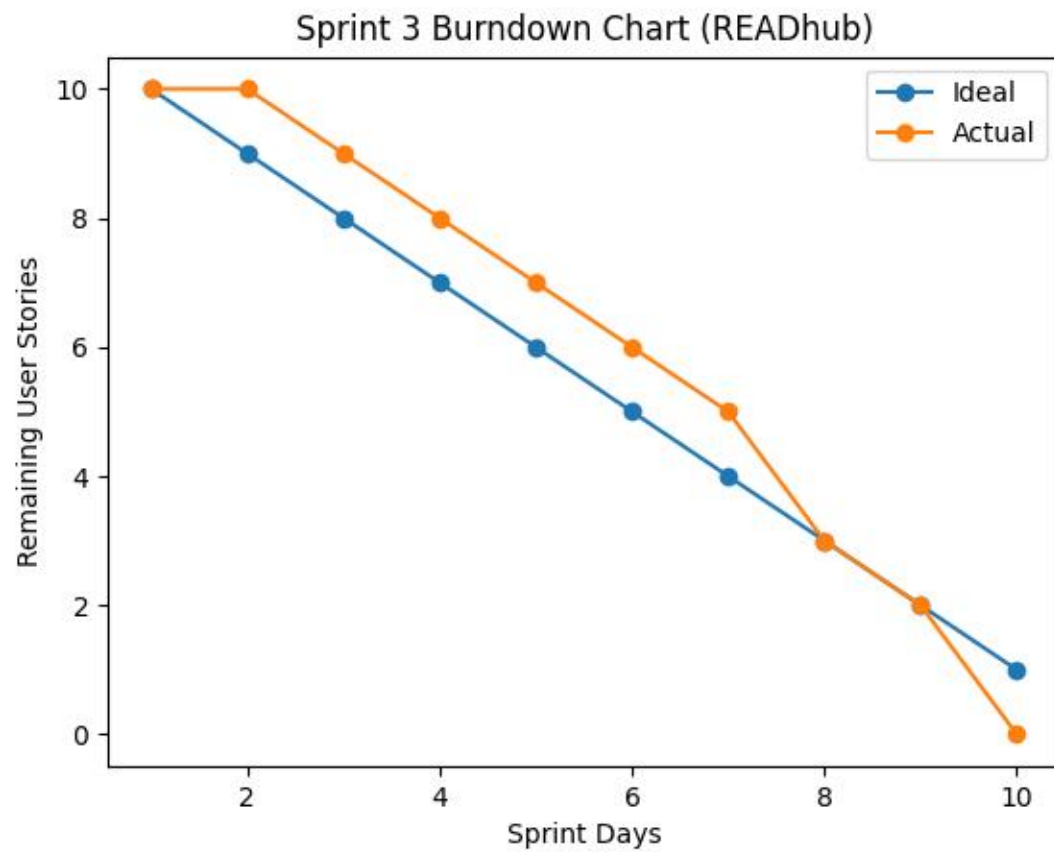
Review and manage book requests from users.

| ID | USER                                       | BOOK TITLE | AUTHOR | NOTES | DATE       | STATUS   | ACTION      |
|----|--------------------------------------------|------------|--------|-------|------------|----------|-------------|
| #3 | kinza5025@gmail.com<br>kinza5025@gmail.com | dnsj       | fgh    | -     | 25/04/2026 | Reviewed | Ma<br>Penc  |
| #2 | kinza5025@gmail.com<br>kinza5025@gmail.com | gwud       | hquah  | -     | 25/04/2026 | Reviewed | Ma<br>Penc  |
| #1 | kinza5025@gmail.com<br>kinza5025@gmail.com | new        | rjg    | fbhdj | 25/04/2026 | Pending  | Ma<br>Revis |

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## 7. Product Burndown Chart

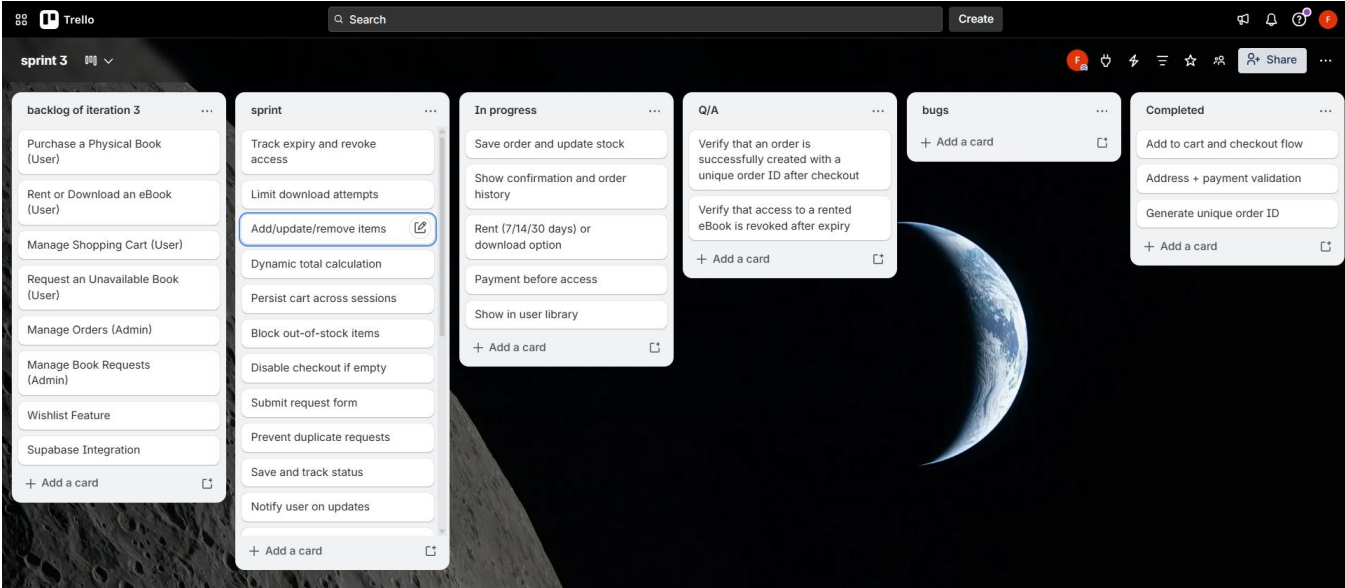
---



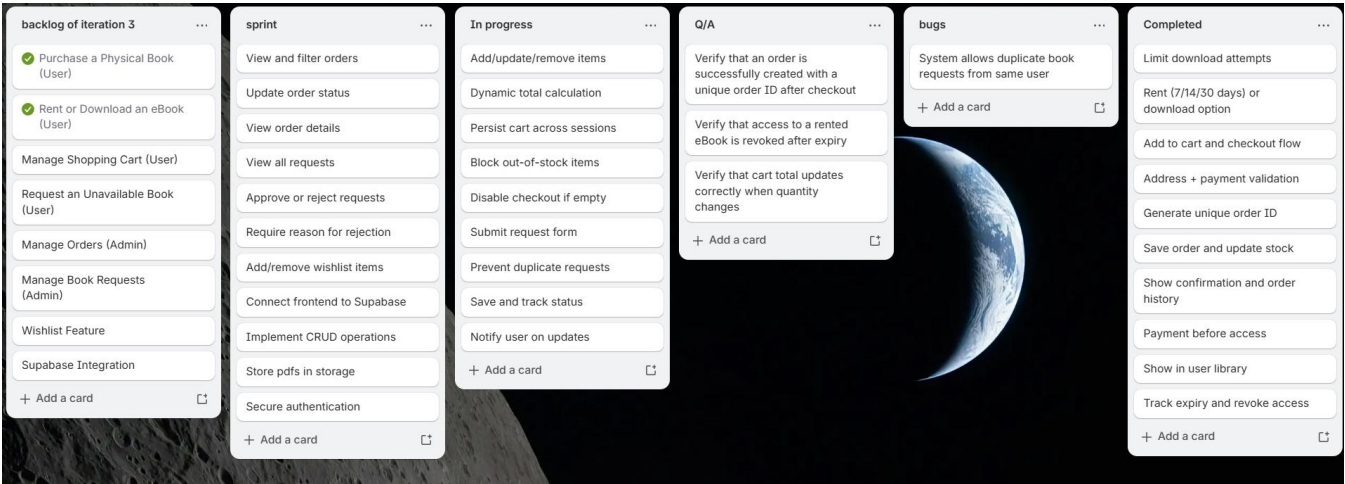
## 8. Trello Board Screenshots

The following Trello board screenshots document the project management workflow across all three sprints.

### Trello Board 1



### Trello Board 2



# Trello Board 3

Trello

Search

Create

sprint 3

backlog of iteration 3

Purchase a Physical Book (User)

Rent or Download an eBook (User)

Manage Shopping Cart (User)

Request an Unavailable Book (User)

Manage Orders (Admin)

Manage Book Requests (Admin)

Wishlist Feature

Supabase Integration

+ Add a card

sprint

+ Add a card

In progress

View and filter orders

Update order status

View order details

View all requests

Approve or reject requests

Require reason for rejection

Add/remove wishlist items

Connect frontend to Supabase

Implement CRUD operations

Store pdfs in storage

Secure authentication

+ Add a card

Q/A

Verify that an order is successfully created with a unique order ID after checkout

Verify that access to a rented eBook is revoked after expiry

Verify that cart total updates correctly when quantity changes

Verify that wishlist items persist after logout/login

+ Add a card

bugs

Wishlist items disappear after user logs back in

error in connecting supabase at first

+ Add a card

Completed

Limit download attempts

Rent (7/14/30 days) or download option

Add to cart and checkout flow

Address + payment validation

Generate unique order ID

Save order and update stock

Show confirmation and order history

Payment before access

Show in user library

Track expiry and revoke access

+ Add a card



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## 9. Boundary Value Analysis Testing

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### Valid Test Cases

| Test Case No. | Test Condition              | Input Email         | Input Password | Expected Result                 | Actual Result    | Accepted |
|---------------|-----------------------------|---------------------|----------------|---------------------------------|------------------|----------|
| 1             | Valid admin login           | admin@gmail.com     | password       | Login successful (Admin access) | Login successful | ✓        |
| 2             | Valid user login (Kinza)    | Kinza5025@gmail.com | password       | Login successful (User access)  | Login successful | ✓        |
| 3             | Valid user login (Fajar)    | ali@gmail.com       | 12345678       | Login successful                | Login successful | ✓        |
| 4             | Valid user login (Fjrowaes) | fjrowaes@gmail.com  | 12345678       | Login successful                | Login successful | ✓        |

### Invalid Test Cases

| Test Case No. | Test Condition       | Input Email         | Input Password | Expected Result | Actual Result        | Accepted |
|---------------|----------------------|---------------------|----------------|-----------------|----------------------|----------|
| 1             | Invalid email format | admin.gmail.com     | 12345678       | Prevent login   | Invalid email format | ✓        |
| 2             | Unregistered email   | unknown@gmail.com   | 12345678       | Prevent login   | User not found       | ✓        |
| 3             | Wrong password       | Kinza5025@gmail.com | 12345          | Prevent login   | Invalid password     | ✓        |

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|    |                         |                      |          |                |                               |   |
|----|-------------------------|----------------------|----------|----------------|-------------------------------|---|
| 4  | Password below boundary | fjrowaes@gmail.com   | abcd     | Prevent login  | Password too short            | ✓ |
| 5  | Empty email             |                      | 12345678 | Prevent login  | Email required                | ✓ |
| 6  | Empty password          | fajarowaes@gmail.com |          | Prevent login  | Password required             | ✓ |
| 7  | Both fields empty       |                      |          | Prevent login  | Missing credentials           | ✓ |
| 8  | Email with spaces       | ali@gmail.com        | zxcv     | Trim or reject | Invalid format / trim failure | ✓ |
| 9  | Wrong admin password    | admin@gmail.com      | asdfg    | Prevent login  | Invalid credentials           | ✓ |
| 10 | Incorrect password      | Kinza5025@gmail.com  | 87654321 | Prevent login  | Invalid password              | ✓ |

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## 10. Work Division

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The following table documents the work division among team members across all three sprints.

| Member | Sprint   | Responsibilities                                                                                                                                                         |
|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kinza  | Sprint 1 | Book catalog UI, Admin book management, Inventory module, Database schema design                                                                                         |
| Kinza  | Sprint 2 | Catalog image management, Database query optimization                                                                                                                    |
| Kinza  | Sprint 3 | Physical book purchase flow, Shopping cart (persistence + UI), Order management admin panel, CO fixes (Inventory, FK Constraint), supabase integration, supabase buckets |
| Urwa   | Sprint 1 | User registration and login pages, Book search and filter, Profile layout                                                                                                |
| Urwa   | Sprint 2 | User profile management, Book image validation, Category management, Rating display fix                                                                                  |
| Urwa   | Sprint 3 | eBook rent/download UI, Checkout address & payment form, Admin order status updates, Image validation CO fix, Category CO fix                                            |
| Fajar  | Sprint 1 | Book detail page, Review display, Navigation components                                                                                                                  |
| Fajar  | Sprint 2 | Rating system, ui management                                                                                                                                             |
| Fajar  | Sprint 3 | eBook expiry logic, Book request form & tracking, Admin book request workflow                                                                                            |

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## 11. Lessons Learnt

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### 11.1 Technical Lessons

- Developing complete features such as book purchase, shopping cart, and eBook rental helped us understand how frontend, backend, and database must work together as one system.
- The shopping cart and order system taught us the importance of maintaining accurate data, especially when handling quantities, totals, and inventory updates at the same time.
- Implementing the eBook rental system introduced time-based logic (expiry and access control), which required careful backend handling.
- Features like wishlist and book requests showed the need for well-structured database design and efficient data operations.
- Using Supabase helped us learn how to manage authentication, storage, and database operations in a secure and scalable way.

### 11.2 Process Lesson

- Sprint 3 included many major features, so progress was slower at the beginning due to dependencies between modules.
- Once core parts (authentication and database structure) were stable, development became faster and more efficient.
- Dividing user stories into smaller tasks made it easier to track progress and allowed team members to work in parallel.
- Carryover tasks from previous sprints increased workload, showing the importance of better time estimation and planning.
- Overall, the sprint followed a pattern of slow start and fast completion, which reflects real-world development.

### 11.3 Collaboration Lessons

- Clear division of work among team members helped avoid confusion and improved productivity.
- Working on shared resources like the database required coordination to prevent conflicts and errors.
- Regular communication helped resolve issues quickly, especially during integration of different modules.
- Reviewing each other's work improved code quality and ensured all features worked correctly together.

### 11.4 Overall Reflection

Sprint 3 was the most important phase of the project, as it involved completing all major features including purchasing, eBook access, cart management, admin functionalities, and user interaction features. Although the initial phase required more time due to system integration, the team was able to complete all tasks successfully by the end of the sprint. This experience highlighted the importance of planning, teamwork, and building systems in a modular way.

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